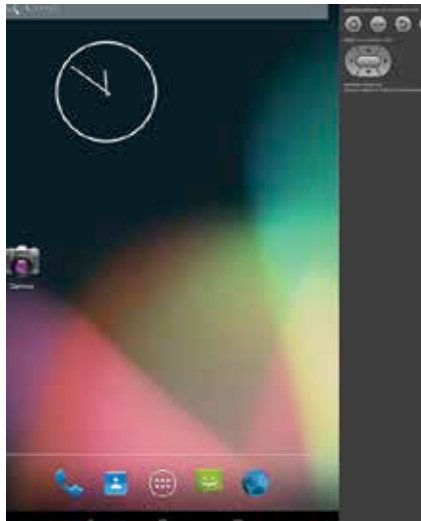


another hard drive, and your problem is solved.

If you are looking for something similar on Linux, know that it is called LVM and has been around for much much longer (early 2000s). A newer and different approach is found in btrfs. The concept itself is even older.

You will note from all these examples though that whatever virtual resource we create, it is backed by a real resource of the same kind. For instance, the virtual processors that appear due to hyper-threading are backed by a real



Android emulator running Android 5.

processor, virtual memory is backed by real memory, and virtual storage is backed by real storage. Herein lies the difference between emulation and virtualisation. In fact you may have heard of hardware assisted virtualisation, a feature present in recent processors that enhances the performance of virtualisation software.

You can virtualise the hardware that you have on your system so that it can be accessed by multiple OSs, for example Windows running directly on your hardware and Linux running in a virtualised machine. In this case Linux could run directly on the host machine. In the case of virtual hyper-threading processors, the instructions they run are just passed to the host hardware, and with Windows storage spaces the data is stored on physical disks.

Let's contrast this to emulation, which is about creating a system that behaves like what it is emulating without the actual presence of such a system. For example a CD / DVD drive emulator does not require an actual CD / DVD drive, it reads data from an image instead and presents a virtual drive that behaves like the real thing.

When you run Android (for ARM) on your computer you are using an emulator, since there is no ARM chip in your computer, instead the QEMU software is emulating the presence of one.